

14. Challenges Faced and Resolution

14.1 Overview

During the execution of the ReqRes API Testing project, several practical challenges were encountered while configuring Postman, managing environment variables, and executing collections through Newman. These issues are common in real-world API testing projects and provided valuable learning opportunities.

This section describes the challenges faced, their root causes, impacts on testing, and the solutions implemented to resolve them.

14.2 Base URL Configuration Issue

Challenge Description

One of the first issues encountered during testing was related to incorrect Base URL configuration in Postman.

Initially, some API requests were created using incomplete or incorrect URLs, which resulted in request failures.

For example:

Incorrect URL:

```
api/login
```

or

```
req/api/users
```

Instead of:

```
https://reqres.in/api/login
```

or

```
{{base_url}}/login
```

Problem Observed

The API requests failed and returned errors such as:

404 Not Found

or

Invalid URL

As a result:

- API requests could not reach the server.
- Test cases failed.
- Collection execution was interrupted.

Root Cause Analysis

The issue occurred because:

- Base URL was not properly configured.
- Requests were created before environment variables were set.
- Hardcoded URLs were used inconsistently across requests.

Impact on Testing

The following impacts were observed:

- Login API execution failed.
- User retrieval APIs returned errors.
- Collection Runner execution produced failures.
- Additional troubleshooting time was required.

Resolution Implemented

A dedicated environment variable was created in Postman:

Variable Name	Value
base_url	https://reqres.in/api

All requests were updated to use:

```
{{base_url}}
```

Example:

```
{{base_url}}/login  
{{base_url}}/users  
{{base_url}}/users/2
```

Result After Resolution

After updating all requests:

- URLs resolved correctly.
 - APIs connected successfully to the ReqRes server.
 - Collection execution completed without URL-related failures.
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Lesson Learned

Using environment variables instead of hardcoded URLs improves:

- Maintainability
- Reusability
- Scalability

This is considered a best practice in API testing projects.

14.3 Environment Variable Issue

Challenge Description

Another challenge encountered was related to Postman Environment Variables not being recognized during execution.

Some requests displayed:

```
{{base_url}}
```

instead of replacing it with the actual URL.

Problem Observed

Requests were executed with unresolved variables, resulting in:

```
Error: Invalid URL
```

or

Could not send request

Root Cause Analysis

Investigation revealed that:

- Environment was created but not selected.
- Variable names were mismatched.
- Some requests referenced incorrect variable names.

Example:

Environment Variable:

```
base_url
```

Request:

```
{{baseurl}}
```

Notice the missing underscore.

Impact on Testing

The issue caused:

- Request execution failures.
 - Collection Runner interruptions.
 - Failed API validations.
-

Resolution Implemented

The following actions were taken:

Step 1: Verify Environment

Created environment:

```
ReqRes Environment
```

Step 2: Configure Variable

```
base_url = https://reqres.in/api
```

Step 3: Select Environment

Before execution:

Top Right Corner → Select ReqRes Environment

Step 4: Verify Variable Names

Ensured consistency:

Correct:

```
{{base_url}}
```

Incorrect:

```
{{baseurl}}
```

Result After Resolution

After correcting environment settings:

- Variables resolved successfully.
 - APIs executed correctly.
 - Collection Runner execution became stable.
-

Lesson Learned

Environment variables must always be:

- Properly configured
- Properly selected
- Consistently referenced

This avoids execution failures and improves test maintainability.

14.4 Newman Execution Issue

Challenge Description

While automating API execution using Newman, installation and execution-related issues were encountered.

Initially, Newman commands failed to execute properly.

Problem Observed

The following error appeared:

```
'newman' is not recognized as an internal or external command
```

In some cases:

```
Command not found
```

was displayed.

Root Cause Analysis

The issue occurred because:

- Newman was not installed globally.
 - Node.js installation was incomplete.
 - PATH environment variables were not updated.
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Impact on Testing

As a result:

- Automated execution could not start.
 - Reports could not be generated.
 - Newman validation could not be performed.
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Resolution Implemented

Step 1: Verify Node.js Installation

Command:

```
node -v
```

Verified Node.js was installed successfully.

Step 2: Install Newman

Command:

```
npm install -g newman
```

Step 3: Verify Installation

Command:

```
newman -v
```

Expected Output:

```
6.x.x
```

Step 4: Execute Collection Again

Command:

```
newman run ReqRes_API_Testing.postman_collection.json
```

Execution completed successfully.

Additional Issue

At one stage, Newman could not locate the collection file.

Error:

```
ENOENT: no such file or directory
```

Resolution

Provided correct file path:

```
newman run "D:\Project\ReqRes_API_Testing.postman_collection.json"
```

Result After Resolution

After configuration:

- Newman executed successfully.
- Collections ran from the command line.
- Test reports were generated successfully.

Lesson Learned

Before executing Newman:

- Verify Node.js installation.
- Install Newman globally.
- Confirm collection file path.
- Validate environment file availability.

These checks help prevent execution failures.

14.5 Challenges Summary

Challenge	Root Cause	Resolution
Base URL Configuration Issue	Incorrect or incomplete URLs	Configured <code>base_url</code> environment variable
Environment Variable Issue	Environment not selected or variable mismatch	Corrected variable names and selected environment
Newman Execution Issue	Newman/Node.js not installed correctly	Installed and configured Newman properly

14.6 Overall Learning from Challenges

The challenges encountered during this project provided practical exposure to common issues faced in API testing projects. Resolving these issues improved understanding of:

- Postman environment management
- API request configuration
- Collection execution workflows
- Newman automation setup
- Troubleshooting techniques

These learnings are directly applicable to real-world API testing and automation projects.
